

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

1st Semester of M. Pharm [Pharmaceutical Technology / Pharmaceutics (Drug Regulatory Affairs)]

University Theory Examination December 2015

PH819.02 Biopharmaceutics and Pharmacokinetics

Date: 12.12.2015, Saturday

Time: 10:00 a.m to 1:00 p.m

Maximum Marks: 70

Instructions:

1. Q.1 and Q.4 are compulsory.
2. Figures to right indicate marks.
3. Section wise separate answer book to be used.
4. Draw neat sketches wherever necessary.

SECTION-I

- Q.1 A) Discuss various pharmaceutical factors affecting drug absorption. 05
- B) Explain how solubility and permeability of drug is determined. 05

- Q.2 A) Explain how fast and fed state BA studies are performed. 05
- B) Describe various methods for calculation of AUC. 05

OR

- Q.2 A) Discuss various types of study designs used in BA-BE study. 05
- B) Write a detailed note on Biopharmaceutical Classification System (BCS). 05

- Q.3 Answer the following (Any three) 15
- (A) Discuss various pharmacokinetic methods used for measurement of bioavailability of drugs.
- (B) Write a brief note on various factors affecting excretion of drugs.
- (C) Explain Volume of Distribution and Plasma Protein Binding. Discuss their relationship with example.
- (D) Briefly describe the mechanism of Renal excretion and concept of clearance.

SECTION-II

- Q.4 What do you mean by IVIVC? Explain the different levels for establishing IVIVC. Write about the applications of IVIVC. 10

Q.5 Discuss about the causes and changes in pharmacokinetic parameters when non-linearity is observed in drug absorption, distribution, metabolism and excretion. **10**

OR

Q.5 Explain one compartment open model for IV bolus administration. Explain how elimination rate constant (K_E) and Elimination Half-life for one compartment open model for IV bolus administration is determined. **10**

Q.6 Answer the following (Any three) **15**

- (A) Write a short note on: Biowaivers in IVIVC.
 - (B) Explain the dose adjustment in 'Renal failure.'
 - (C) Draw a labeled diagram of a typical plasma concentration-time profile curve and explain in brief.
 - (D) Explain any one method for the calculation of absorption rate constant.
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